

THALLIUM

(Data in kilograms unless otherwise noted)

Domestic Production and Use: Thallium has not been recovered in the United States since 1981. Consumption of thallium metal and thallium compounds was valued at \$258,600. The primary end uses included the following: radioactive thallium-201 used for medical purposes in cardiovascular imaging; thallium as an activator (sodium iodide crystal doped with thallium) in gamma radiation detection equipment (scintillometer); thallium-barium-calcium-copper oxide high-temperature superconductor used in filters for wireless communications; thallium in lenses, prisms, and windows for infrared detection and transmission equipment; thallium-arsenic-selenium crystal filters for light diffraction in acousto-optical measuring devices; and thallium in mercury alloys for low-temperature measurements. Other uses include: as an additive in glass to increase its refractive index and density, a catalyst for organic compound synthesis, and a component in high-density liquids for gravity separation of minerals.

Salient Statistics—United States:	2014	2015	2016	2017	2018^e
Production, refinery	—	—	—	—	—
Imports for consumption:					
Unwrought metal and metal powders	44	—	—	—	—
Waste and scrap	—	—	—	—	23
Other articles	53	334	193	—	41
Exports:					
Unwrought metal and metal powders	51	104	56	34	100
Waste and scrap	1,430	1,450	286	364	150
Other articles	1,050	1,070	973	1,560	2,000
Consumption, estimated ¹	97	334	193	—	64
Price, metal, dollars per kilogram ²	7,200	7,400	7,400	4,200	4,000
Net import reliance ³ as a percentage of estimated consumption	NA	NA	NA	NA	NA

Recycling: None.

Import Sources (2014–17): United Kingdom, 100%.

Tariff: Item	Number	Normal Trade Relations 12–31–18
Unwrought and powders	8112.51.0000	4.0% ad val.
Waste and scrap	8112.52.0000	Free.
Other	8112.59.0000	4.0% ad val.

Depletion Allowance: 14% (Domestic and foreign).

Government Stockpile: None.

Events, Trends, and Issues: In 2018, the price for thallium metal declined for the second year by 5%, likely owing to decreased use and domestic availability from scrap and waste. In 2018, China maintained its policy of eliminating toll-trading tax benefits on exports of thallium that began in 2006, thus contributing to the reduced supply to markets outside of China. In September 2018, the United States Trade Representative released a list of approximately \$200 billion worth of Chinese imports that would be subject to an additional 10% tariff beginning in late September. Further tariffs of 25% will be imposed in January of 2019. Thallium (unwrought), powders, waste and scrap, and other articles were included in the list of Chinese imports that will be subjected to additional tariffs.

Demand for thallium for use in cardiovascular-imaging applications has declined, owing to superior performance and availability of alternatives, such as the medical isotope technetium-99. A global shortage of technetium-99 from 2009 to 2011 had contributed to an increase in thallium consumption during that time period. Since 2011, consumption of thallium has declined significantly. Small quantities of thallium are used for research.

THALLIUM

Two of the leading global markets for thallium were glass lenses, prisms, and windows for fiber optics, and optics for digital cameras. The majority of producers of these products were in China, Japan, and the Republic of Korea. A search of the U.S. Patent and Trademark office shows that patents were filed in 2018 to use thallium as a metal plating enhancer in batteries and as a variable resistance layer in memory devices to increase thermal and electrical conductivity. Other applications were for antibody detection labels with radioactive isotopes for research, and for quantum dots and solder.

Thallium metal and its compounds are highly toxic materials and are strictly controlled to prevent harm to humans and the environment. Thallium and its compounds can be absorbed into the human body by skin contact, ingestion, or inhalation of dust or fumes. The leading sources of thallium released into the environment are coal-burning powerplants and smelters of copper, lead, and zinc ores. The major sources of thallium in drinking water are ore-processing sites and discharges from drugs, electronics, and glass factories. Under its national primary drinking water regulations for public water supplies, the U.S. Environmental Protection Agency has set an enforceable Maximum Contaminant Level of 2 parts per billion of thallium in drinking water.

World Refinery Production and Reserves:⁴ Thallium is produced commercially in only a few countries as a byproduct in the roasting of copper, lead, and zinc ores and is recovered from flue dust. Because most producers withhold thallium production data, global production data are limited. In 2018, global production of thallium was estimated to be less than 8,000 kilograms. China, Kazakhstan, and Russia were believed to be leading producers of primary thallium. Since 2005, substantial thallium-rich deposits have been identified in Brazil, China, Macedonia, and Russia.

World Resources: Although thallium is reasonably abundant in the Earth's crust, estimated at about 0.7 part per million, it exists mostly in association with potassium minerals in clays, granites, and soils, and it is not generally considered to be commercially recoverable from those materials. The major source of recoverable thallium is the trace amounts found in copper, lead, zinc, and other sulfide ores. Quantitative estimates of reserves are not available, owing to the difficulty in identifying deposits where thallium can be extracted economically. Previous estimates of reserves were based on the thallium content of zinc ores. World resources of thallium contained in zinc resources could be as much as 17 million kilograms; most are in Canada, Europe, and the United States. Global resources of coal contain an estimated 630 million kilograms of thallium.

Substitutes: Although other materials and formulations can substitute for thallium in gamma radiation detection equipment and optics used for infrared detection and transmission, thallium materials are presently superior and more cost effective for these very specialized uses. The medical isotope technetium-99 can be used in cardiovascular-imaging applications instead of thallium.

Nonpoisonous substitutes, such as tungsten compounds, are being marketed as substitutes for thallium in high-density liquids for gravity separation of minerals.

⁰Estimated. NA Not available. — Zero.

¹Estimated to be equal to imports.

²Estimated price of 99.99%-pure granules or rods in 100- to 250-gram or larger lots.

³Defined as imports – exports. Consumption and exports of unwrought thallium were from imported material or from a drawdown in unreported inventories.

⁴See Appendix C for resource and reserve definitions and information concerning data sources.